VocalIQ

AI-Powered Speaking Coach

Project Proposal for CS651: Web Systems

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CS651: Web Systems

1. Executive Summary

VocalIQ is an AI-powered web application that acts as a personal speaking coach. Users upload or record a video of themselves speaking, and the system analyzes their facial expressions, voice tone, speech content, and delivery — then provides a ranked evaluation, timestamped coaching feedback, and a two-pass AI-generated improvement plan. The system is designed to help students, professionals, and public speakers become more confident, persuasive, and audience-aware communicators.

2. Problem Statement

Public speaking is consistently ranked as one of the top fears and professional weaknesses across demographics. Despite its importance:

- Human speaking coaches cost $100–300/hour and are inaccessible to most students
- Existing tools (Grammarly, Otter.ai) analyze text only — not delivery, emotion, or vocal presence
- There is no affordable, on-demand tool that watches you speak and gives real coaching feedback
- Speakers have no objective data on nervousness, filler words, pacing, or emotional tone

VocalIQ fills this gap with AI-driven, real-time, multi-modal coaching accessible entirely through a browser.

3. Proposed Solution

VocalIQ allows users to:

1. Record themselves speaking directly in the browser OR upload an existing video
2. Optionally provide context: scenario (pitch, interview, presentation), audience (investors, professors, general public), and goal (persuade, inform, inspire)
3. Receive a structured coaching report including:
    - Language detection
    - Overall score (1–10) with justification
    - Auto-generated captions with timestamps
    - Timestamped findings: nervousness, stuttering, filler words, eye contact breaks, enthusiasm, professional tone
    - Specific fix for each timestamped issue
    - A two-pass AI coaching chain: initial assessment then deep coaching refinement
    - Vocal control tips (pacing, tone, pausing, projection)

4. Core Features

4.1 Video Input (Record or Upload)

Users can record a video directly from their browser camera using the MediaRecorder API, or upload a pre-recorded video file. Both paths feed into the same analysis pipeline. An audio-only fallback mode is also supported.

4.2 Language and Context Detection

Gemini automatically detects the spoken language and identifies the speech context. Users can optionally provide scenario context which is injected into the Gemini prompt to make coaching context-aware.

4.3 Speech Score (1–10 Ranking)

The system returns a structured overall score from 1–10 with a breakdown across dimensions: content clarity, emotional delivery, vocal confidence, pacing, and audience engagement.

4.4 Timestamped Caption Analysis

The system generates a full transcript with timestamps. Each segment is tagged with detected issues:

- Nervousness (voice drop, hesitation)
- Stuttering or repetition
- Filler words ("um", "uh", "like", "you know")
- Eye contact breaks
- Monotone delivery
- Enthusiasm or strong delivery moments
- Professional tone segments

Each flagged segment includes a specific, actionable fix.

4.5 Two-Pass Gemini Coaching Chain

Pass 1: Gemini analyzes the video and returns an assessment — language, score, top improvement tips, and timestamped transcript analysis.

Pass 2: The output of Pass 1 is fed back into Gemini as context, generating a deeper coaching report with detailed exercises, rephrasing examples, vocal control techniques, and a practice plan.

4.6 Vocal Control Tips

The final report includes a dedicated section on vocal mechanics: pacing guidance, pause technique for filler words, breath control for nervousness, and modulation tips for monotone delivery.

4.7 Session History and Progress Tracking

All coaching sessions are saved per user. Users can view past sessions, track their score over time on a progress chart, and compare transcripts across sessions.

4.8 With or Without Video Mode

- Full mode: video (face + voice) — facial expression + voice analysis
- Audio-only mode: voice only — transcript and language analysis without camera

5. Technical Architecture

5.1 Technology Stack

Layer	Technology
Frontend	React (Vite), Tailwind CSS, Recharts, Framer Motion
Backend	Node.js, Express
AI Engine	Google Gemini 2.5 Flash (multimodal: video, audio, text)
Storage	Google Cloud Storage (video), Google Firestore (sessions)
Authentication	Google OAuth 2.0 + JWT
Video Processing	Gemini File API (supports video up to 2 GB)

5.2 Key Technical Components

Gemini File API:

Videos are uploaded to Gemini's File API (not as base64 inline data, which is unsuitable for large files). Gemini processes the video server-side and returns a fileUri used in subsequent analysis calls. Files auto-delete from Gemini after 48 hours; permanent copies are retained in GCS.

Two-Pass Prompt Chain:

Pass 1 requests: language, score, transcript array with timestamps and issue tags, and top improvement tips. Pass 2 receives the Pass 1 JSON output as context and generates deep coaching content for each identified issue.

Timestamped Transcript Data Shape:

```
{ start: "0:23", end: "0:31", text: "...", issues: ["nervousness"], fix: "..." }
```

5.3 Firestore Data Model

Collection: coaching_sessions

Fields:

- userId, videoUrl, language, overallScore
- scoreBreakdown (object)
- transcript (array)
- improvementTips (array)
- detailedCoaching (object)
- vocalControlTips (object)
- createdAt, sessionDurationSeconds, mode

5.4 API Endpoints

```
POST  /api/coaching/upload    — accepts video/audio, uploads to Gemini File API +
GCS
POST  /api/coaching/analyze   — runs two-pass Gemini chain, returns full coaching
report
GET   /api/coaching/sessions  — returns user session history
GET   /api/coaching/sessions/:id — returns full report for one session
```

6. User Interface

Page 1 — Session Setup

- Select mode: Record video / Upload video / Audio only
- Optional context: Scenario, Audience, Goal
- "Start Analysis" button

Page 2 — Recording / Upload

- Live camera preview with record button
- OR drag-and-drop video file upload
- Progress bar during processing (30–90 seconds)

Page 3 — Coaching Report

- Tab 1: Overview — Score (1–10), language, breakdown chart
- Tab 2: Transcript and Timestamps — interactive timeline, color-coded by issue
- Tab 3: Improvement Tips — two-pass coaching with exercises and examples
- Tab 4: Vocal Control — pacing, pause technique, breath control

Page 4 — Progress Dashboard

- Line chart of score over time
- Session history list
- Side-by-side session comparison view

7. Monetization Strategy

Pricing Tiers

Tier	Price	Sessions	Features
Free	$0 / month	3 / month	Audio-only mode, basic score
Pro	$12.99 / month	Unlimited	Video mode, full timestamped report, progress tracking
Teams	$49 / month	Unlimited	Admin dashboard, aggregate team scores

Target Markets

- University students: interview prep, class presentations, debate practice
- Professionals: pitch practice, sales training, conference talks
- Corporate L&D teams: employee communication training programs

Competitive Advantage

No existing tool combines video emotion analysis + timestamped transcript coaching + two-pass AI refinement in a single browser-based web app at this price point.

8. Development Timeline

Week	Deliverables
Week 1	Gemini File API integration, video upload and recording, Pass 1 prompt and structured JSON output
Week 2	Pass 2 coaching chain, timestamped transcript analysis, coaching report UI
Week 3	Progress dashboard, session history, score tracking chart, audio-only fallback
Week 4	Polish, edge case handling, deployment to Google Cloud Run, end-to-end testing

9. Risks and Mitigations

Risk	Mitigation
Video processing latency (30-90 seconds)	Clear progress indicator with step-by-step status, async processing with polling endpoint, optimistic UI patterns
Large video file sizes	Client-side compression before upload, 10-minute session cap enforced on frontend, Gemini File API supports files up to 2 GB
Gemini API cost at scale	Cache all reports in Firestore so the same video is never re-analyzed; rate-limit free tier users to 3 sessions/month
Accuracy of emotion/nervousness detection	Frame outputs as 'AI coaching suggestions, not clinical assessment' — user controls context and can dismiss any suggestion

10. Conclusion

VocalIQ builds directly on the multimodal AI infrastructure established in AuraBoard and extends it into a commercially viable, educationally valuable product. The two-pass Gemini chain, timestamped coaching feedback, and context-aware analysis represent a meaningful technical contribution. The system addresses a real, underserved market need: accessible, on-demand speaking coaching for the modern professional.